

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What is the **cytologic hallmark** of an adenomatous polyp?
2. Which polyp type has a **sawtooth** architecture limited to the upper half of the crypts?
3. Name the **triad** of solitary rectal ulcer syndrome.

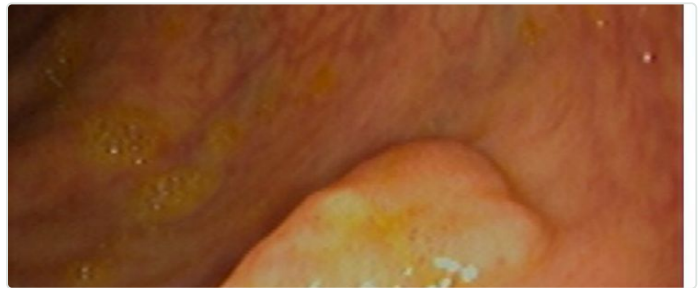
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Intestinal polyps — the frame

Where	Most common in the colon .
Gross	Sessile = without a stalk; pedunculated = with a stalk.
Classification	Non-neoplastic : inflammatory, hamartomatous, hyperplastic. Neoplastic : the commonest is the adenomatous polyp .



Sessile polyp (no stalk)



Pedunculated polyp (stalk)

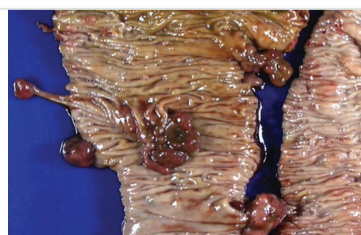
2 Non-neoplastic polyps — one identifier each

Hamartoma = disorganised, tumour-like growth of mature cell types normally present at the site; single or multiple; sporadic or syndromic.

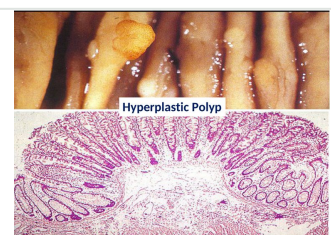
Polyp	Setting	The identifying feature
Inflammatory (solitary rectal ulcer syndrome)	Triad: rectal bleeding, mucus discharge, inflammatory lesion of the anterior rectal wall . Due to impaired relaxation of the anorectal sphincter	Chronic cycles of injury and healing produce a polypoid mass
Juvenile (hamartomatous)	Commonest hamartomatous polyp ; in the rectum ; rectal bleeding and prolapse through the anal sphincter. Sporadic, or (per the slide) part of Peutz-Jeghers syndrome	Cystically dilated crypts filled with mucin and inflammatory debris; no dysplasia
Peutz-Jeghers (hamartomatous)	Rare autosomal dominant : multiple GIT hamartomatous polyps (most common in the small intestine) + mucocutaneous hyperpigmentation + increased risk of malignancy (colon, pancreas, breast, lung, ovaries, uterus, testes)	Arborizing (tree-like) network of connective tissue, smooth muscle, lamina propria and glands lined by normal-appearing epithelium
Hyperplastic	Commonest polyp of the adult colorectum ; more in the colon; 6th–7th decades	Serrated, sawtooth architecture limited to the upper half of crypts ; no conventional dysplasia. Risk of progression to malignancy is questionable



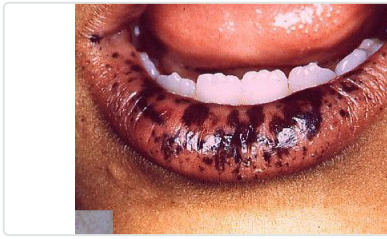
Solitary rectal ulcer with polyp



Juvenile polyps (gross)



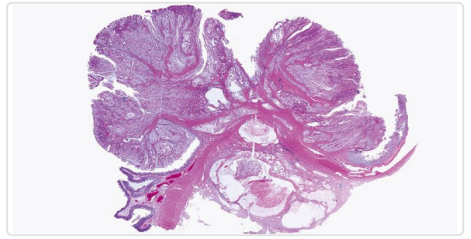
Hyperplastic polyp



Peutz-Jeghers: mucocutaneous hyperpigmentation



Peutz-Jeghers: hamartomatous polyps



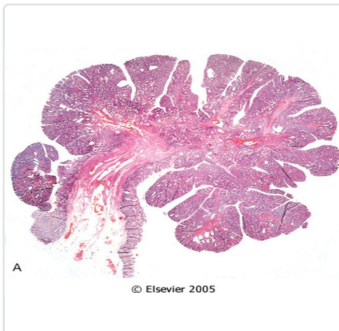
Peutz-Jeghers: arborizing histology

3 Adenomatous polyps (adenomas)

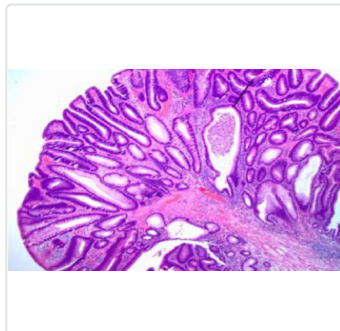
THE BLOCK'S RECURRING PATTERN

Chronic injury → metaplasia → **dysplasia** → carcinoma. In the colon: **adenoma (epithelial dysplasia) → colorectal adenocarcinoma**. Adenomas give rise to the majority of colorectal adenocarcinomas, but not all adenomas progress.

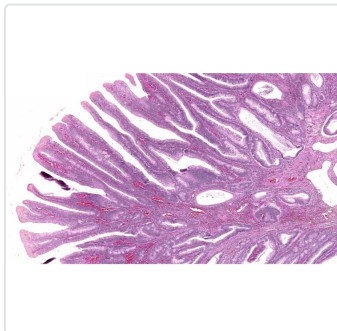
What	Neoplastic polyps; the cytologic hallmark is epithelial dysplasia . Benign.		
Who / size	Beginning at age 50 . Size 0.3 to 10 cm; pedunculated or sessile.		
	Tubular	Villous	Tubulovillous
Shape	Tend to be small, pedunculated	Often larger and sessile	Mixture of tubular and villous elements
Architecture	Small rounded or tubular glands	Covered by slender villi	Mixed
Cancer risk	—	More frequently contain foci of invasive malignancy	—



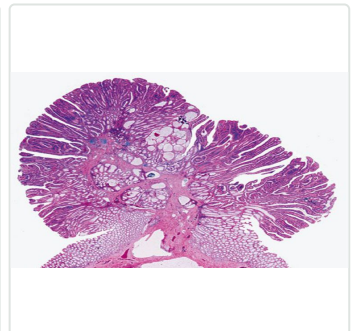
Pedunculated adenomatous polyp



Tubular adenoma



Villous adenoma



Tubulovillous adenoma

P Must-remember pearls

- **Polyps are commonest in the colon**; adenomatous polyps are the commonest neoplastic polyp.
- **Juvenile = rectum, bleeding and prolapse, cystically dilated mucin-filled crypts, no dysplasia.**
- **Peutz-Jeghers = autosomal dominant + hamartomatous polyps (small bowel) + mucocutaneous pigmentation + cancer risk**; tree-like arborizing histology.
- **Hyperplastic = commonest in adult colorectum; sawtooth in the upper half of crypts; no dysplasia.**
- **Adenomas are defined by dysplasia**; villous adenomas are larger, sessile, and more often contain invasive malignancy.

Q Test yourself

1. A 5-year-old has rectal bleeding and a polyp that protrudes through the anal sphincter. Histology shows cystically dilated crypts filled with mucin and inflammatory debris, with no dysplasia. The polyp is:

- A juvenile (hamartomatous) polyp
- An adenomatous polyp
- A hyperplastic polyp
- An inflammatory polyp of solitary rectal ulcer syndrome
- A villous adenoma

2. A young patient has multiple small-bowel polyps and mucocutaneous hyperpigmentation. Histology shows an arborizing network of smooth muscle, lamina propria and glands lined by normal epithelium. What is the inheritance pattern of this syndrome?

- A. Autosomal recessive
- B. X-linked recessive
- C. Mitochondrial
- D. Autosomal dominant
- E. Sporadic only

3. A colonoscopic biopsy shows serrated, sawtooth glands confined to the upper half of the crypts, without conventional dysplasia. The polyp is:

- A. Tubular adenoma
- B. Villous adenoma
- C. Hyperplastic
- D. Juvenile
- E. Peutz-Jeghers

4. Which polyp is larger, sessile, covered by slender villi and more frequently contains foci of invasive malignancy?

- A. Tubular adenoma
- B. Hyperplastic polyp
- C. Juvenile polyp
- D. Inflammatory polyp
- E. Villous adenoma

5. Which triad characterises solitary rectal ulcer syndrome?

- A. Diarrhea, weight loss, villous atrophy
- B. Rectal bleeding, mucus discharge, inflammatory lesion of the anterior rectal wall
- C. Abdominal pain, vomiting, distension
- D. Mucocutaneous pigmentation, polyps, family history
- E. Tenesmus, fever, pneumaturia

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — Juvenile polyps are the commonest hamartomatous polyp, in the rectum, with bleeding and prolapse and mucin-filled cystic crypts without dysplasia.

2 • D — Peutz-Jeghers syndrome is a rare autosomal dominant disorder with hamartomatous polyps, mucocutaneous hyperpigmentation and increased malignancy risk.

3 • C — Hyperplastic polyps show serrated, sawtooth architecture limited to the upper half of the crypts, with no conventional dysplasia.

4 • E — Villous adenomas are often larger and sessile with slender villi and more often contain invasive malignancy; tubular adenomas are small and pedunculated.

5 • B — The triad is rectal bleeding, mucus discharge and an inflammatory lesion of the anterior rectal wall, from impaired anorectal sphincter relaxation.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL2, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.